

Frontend – Assignment 1

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Q1. HTTP Request–Response Cycle

The HTTP Request–Response Cycle is the process through which a client, usually a web browser, communicates with a web server to obtain a resource such as an HTML page, CSS file, image, or other data.

Major steps: (1) The user enters a URL or clicks a link. (2) The browser establishes a connection and sends an HTTP request. (3) The server receives and processes the request. (4) The server sends an HTTP response containing a status code, headers, and usually a response body. (5) The browser processes the response and renders the web page.

Diagram: User → Browser (Client) → HTTP Request → Web Server → HTTP Response → Browser → Web Page

Q2. Introduction to Web Technologies

1. HTML: HTML (HyperText Markup Language) provides the structure and content of a web page using elements such as headings, paragraphs, links, images, tables, and forms.
2. CSS: CSS (Cascading Style Sheets) controls the presentation of HTML content, including colors, fonts, spacing, borders, layouts, and responsive design.
3. JavaScript: JavaScript adds behavior and interactivity to web pages. It can respond to user actions, modify page content, validate forms, and communicate with servers.
4. Bootstrap: Bootstrap is a frontend framework that provides a responsive grid system, utility classes, reusable components, and ready-made styling for faster interface development.
5. jQuery: jQuery is a JavaScript library that simplified DOM manipulation, event handling, animations, and asynchronous requests.
6. React.js: React.js is a JavaScript library for building user interfaces using reusable components. It helps developers manage complex, dynamic frontend interfaces efficiently.

Q3. Client-Side vs. Server-Side Technologies

Client-side technologies run mainly in the user's browser and are responsible for presentation and interaction. Examples include HTML, CSS, and JavaScript.

Server-side technologies run on a web server and handle application logic, database operations, authentication, and generating responses. Examples include Java, Python, PHP, and Node.js.

Difference: Client-side code executes on the client/browser, while server-side code executes on the server. Client-side technologies mainly control the user interface and interaction; server-side technologies mainly handle data processing and application logic.

Q4. Block-Level Elements in HTML

Block-level elements normally begin on a new line and take up the available horizontal space of their containing element. They are commonly used to structure a page into larger content blocks.

Examples: <div>, <p>, <h1>, <section>, and <header>.

Q5. Semantic HTML Elements

Semantic HTML elements clearly describe the meaning and purpose of their content. They improve code readability, accessibility, maintainability, and understanding of page structure.

Examples include <header>, <nav>, <main>, <section>, <article>, <aside>, and <footer>.

Q6. Creating a Table Using HTML

HTML code:

```
<table border="1">
  <tr><th>Sr. No.</th><th>Name</th><th>Designation</th><th>Department</th></tr>
  <tr><td>1</td><td>John</td><td>Manager</td><td>Sales</td></tr>
  <tr><td>2</td><td>Smith</td><td>Sr. Manager</td><td>Marketing</td></tr>
  <tr><td>3</td><td>Jane</td><td>Manager</td><td>HR</td></tr>
</table>
```

Q7. Creating a Nested List Using HTML

HTML code:

```
<ul>

  <li>River

    <ul>

      <li>Ganga</li>

      <li>Yamuna</li>

      <li>Brahmaputra</li>

      <li>Narmada</li>

    </ul>

  </li>

</ul>
```

Q8. Types of CSS

1. Inline CSS: CSS is written directly inside an HTML element using the style attribute.

Example: <p style="color: blue;">Hello</p>

2. Internal CSS: CSS is written inside a <style> element in the HTML document's <head>.

Example: <style> p { color: blue; } </style>

3. External CSS: CSS is stored in a separate .css file and connected using a <link> element.

Example: <link rel="stylesheet" href="style.css">

Q9. Practical Understanding

HTML – Structure: creates headings, paragraphs, links, images, forms, tables, and the basic structure of a web page.

CSS – Presentation: controls colors, fonts, spacing, layout, appearance, and responsiveness.

JavaScript – Behavior: adds interaction, dynamic changes, validation, events, and client-side logic.

Q10. Short Answer

- a. Bootstrap: Bootstrap helps developers create responsive and consistent interfaces quickly. It provides a grid system, utility classes, and reusable components such as buttons, navigation bars, cards, and forms.
- b. jQuery: jQuery is a JavaScript library that simplified common frontend tasks such as DOM manipulation, event handling, animations, and AJAX requests. It was widely used because it reduced code and helped handle browser differences.
- c. React.js: React.js is a JavaScript library for building component-based user interfaces. It helps manage dynamic and complex interfaces by breaking the UI into reusable components and updating the interface efficiently.